MechRecon (Mechanical Reconstruction)

Reconstructing Grain-Average Stress Fields to Satisfy Equilibrium using Discrete Harmonic Expansions

Purpose: To modify a grain-averaged stress (or elastic strain) field to better satisfy equilibrium within and between grains.

Approach:

- Define a stress distribution over a polycrystal using finite element (FE) interpolation.
- The stress distribution requires that the stress be continuous within grains, but may exhibit jumps across grain boundaries.
- The nodal point values of the FE interpolation are represented by discrete harmonic expansions.
- By adjusting the weights of the expansion coefficients, a grain-averaged field is brought into equilibrium.
- An optimization procedure is used to make the adjustment.
- The objective function in the optimization procedure is based on errors in traction equilibrium at grain interfaces and in local equilibrium over the interior of grains.

Methodology:

- Create/import the input data: meshed instantiation from **Neper** (.msh file); harmonic mode files from **MechMonics**; grain averaged stresses (experimental data from HEXD experiments or synthetic data from finite element simulations; experimental data is in the form of grain-averaged elastic strains, use MechToolBox script to convert to stresses.).
- Execute: **MechRecon().** The program will prompt user for necessary input and output file names.
- **MechRecon** minimizes an objective function based on the equilibrium errors subject to the constraint that grain averaged stresses should equal the values provided in the input data.
- The MatLab fmincon function is called within **MechRecon** to perform the optimization.
- **MechRecon** returns the mode weights and residual errors along with the optimized stress distribution.

Citation:

P. R. Dawson and M. P. Miller. Using Discrete Harmonic Expansions and Equilibrium Conditions to Estimate Intragrain Stress Distributions in Polycrystals from Grain-Averaged Data. arXiv:2106.15769v1 [cond-mat.mtrl-sci], 2021.

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Recall from the **MechMonics** Overview, the representation of field data using a finite element representation takes the form:

$a(\mathbf{x}) = [N(\mathbf{x})]\{A\}$ where $[N(\mathbf{x})]$ are the interpolation functions and $\{A\}$ are the nodal point values.

The nodal point values are approximated with an expansion:

$$\{A\} = \sum_k \beta^k \{U^k\}$$

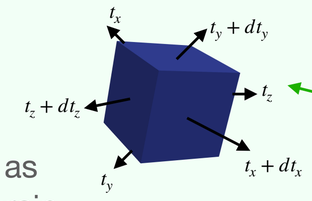
$\{U^k\}$ are the mode shapes already computed in MechMonics. β^k are the weights that will be computed in **MechRecon**.

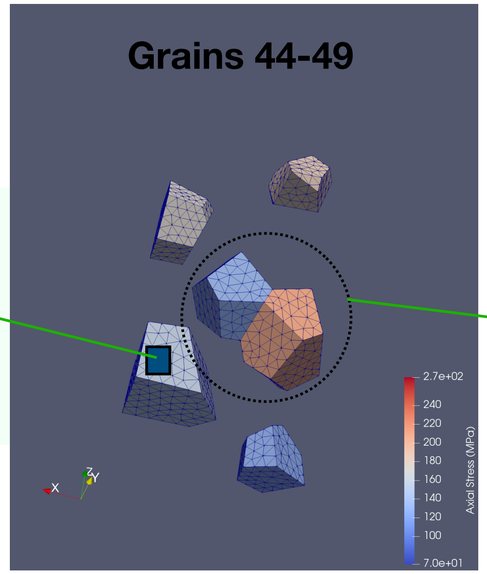
The weights are determined using an optimization procedure.
An objective function is defined :

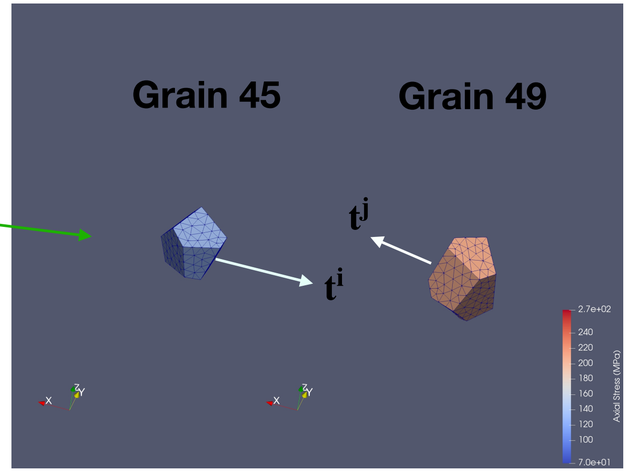
$$F = w^b F^b + w^v F^v$$

Local Equilibrium Objective Function:

$$F^v = \{t^v\}^T\{t^v\}$$
$$\{t^v\} = \{\nabla\sigma\}$$







F^v for all grains is computed as the sum over all elements of a grain using the quadrature point weights and the element volumes as weights.

Traction Objective Function:

$$F^b = \{t^b\}^T\{t^b\}$$
$$\{t^b\} = \{t^i + t^j\}$$

F^b for all grain interfaces is computed as the sum over all adjoining elements at grain interfaces using the interface areas as weights.

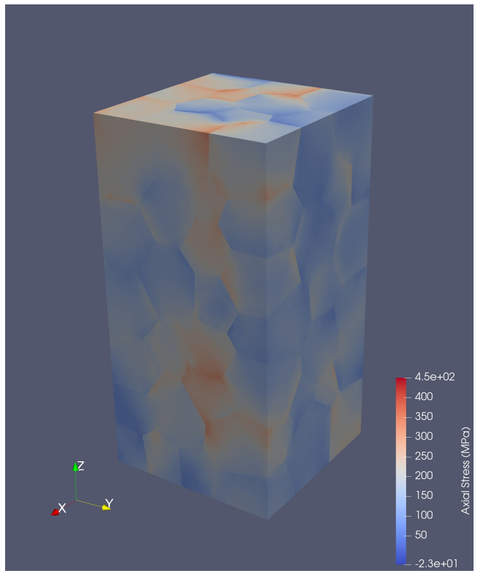
Example — Polycrystalline Tensile Sample

Recovered Axial Stress Distributions

Input: Synthetic HEXD Grain-Averaged Stress Distribution

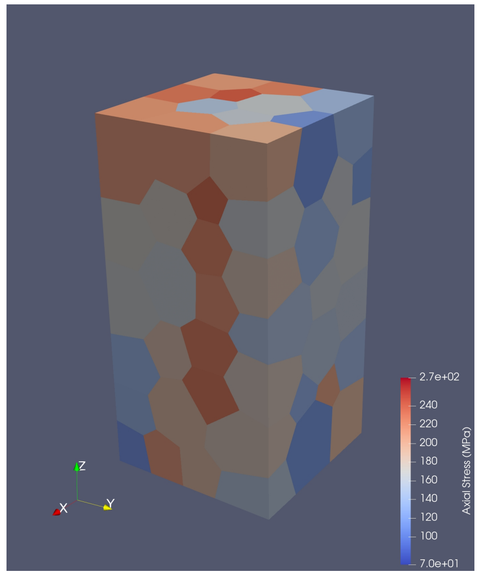
Spatial reconstruction using:

MechMet Solution



Grain-averaged stresses

Synthetic HEXD



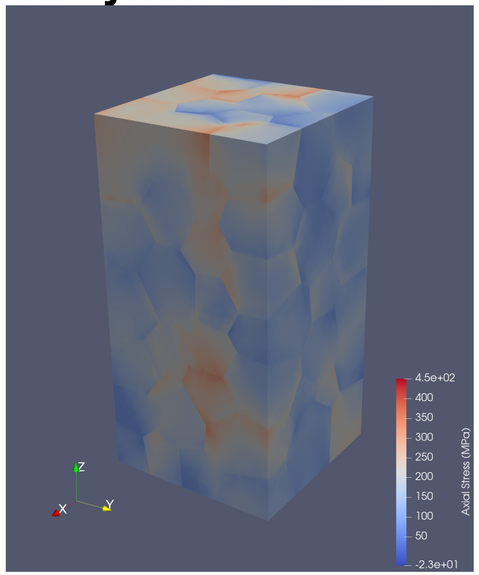
- Harmonic expansions for 6 stress components
- Objective function is a weighted combination of GB traction equilibrium (F^b) and local equilibrium over the grain volume (F^v)
- Equality constraint for HEXD data

Example — Polycrystalline Tensile Sample

Recovered Axial Stress Distributions

Input: Synthetic HEXD Grain-Averaged Stress Distribution

Synthetic HEXD (1 mode fit to the MechMet stress distribution)



For 1Mode , the method returns the data since 1st mode is the average

Spatial reconstruction using:

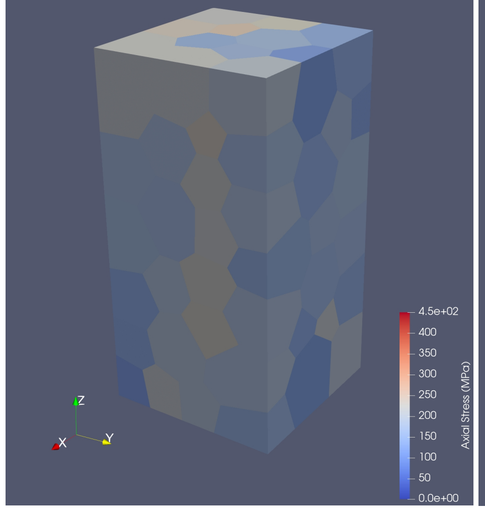
- Harmonic expansions for 6 stress components
- Objective function is a weighted combination of GB traction equilibrium (F^b) and local equilibrium over the grain volume (F^v)
- Equality constraint for HEXD data

$$F = w^b F^b + w^v F^v$$

$$w^b = 0.3$$

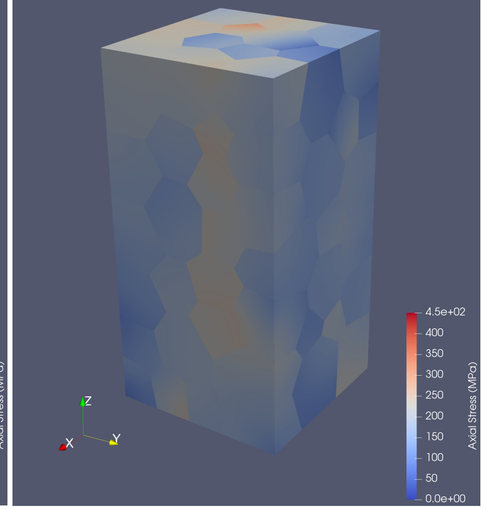
$$w^v = 1.0$$

1 Mode



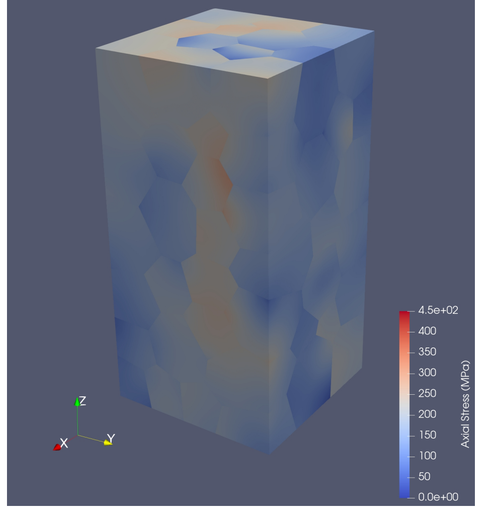
$F_1/F_1 = 1$

4 Modes



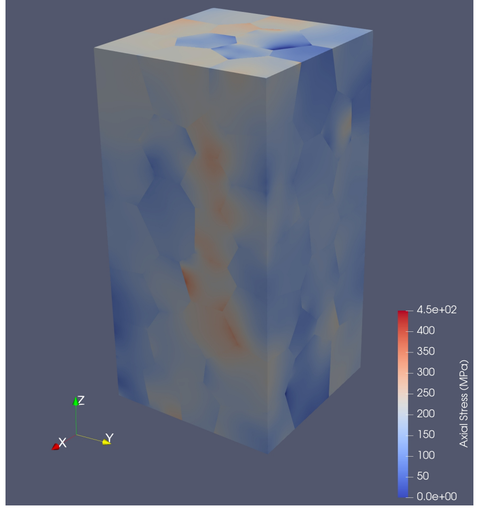
$F_4/F_1 = 0.62$

10 Modes



$F_{10}/F_1 = 0.39$

28 Modes



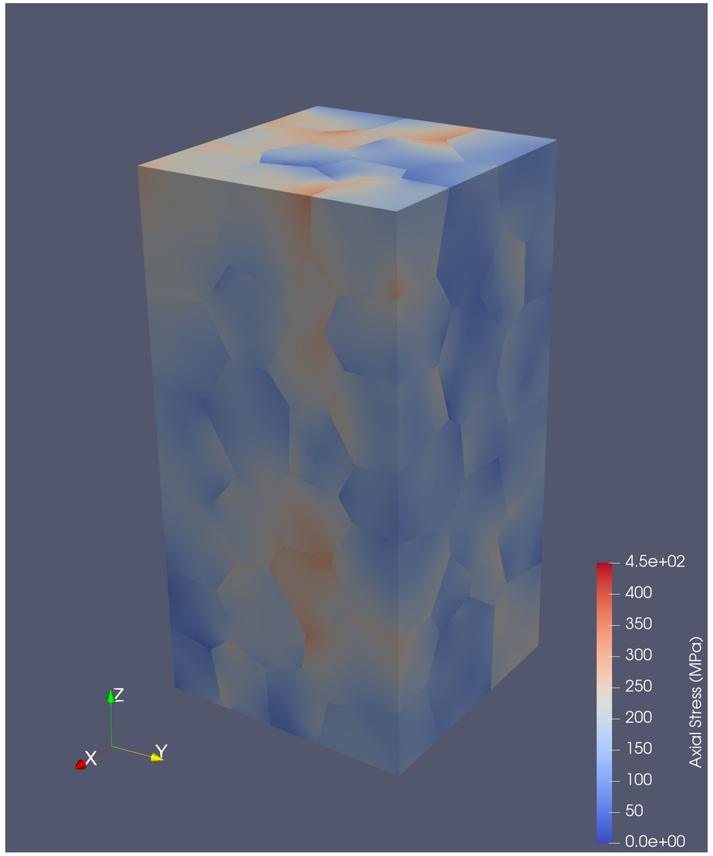
$F_{28}/F_1 = 0.19$

Example — Polycrystalline Tensile Sample

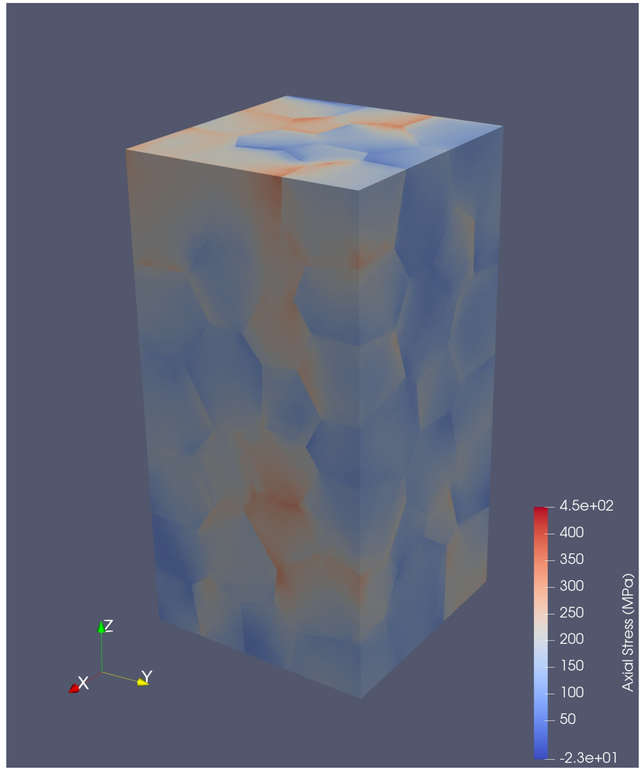
Recovered Axial Stress Distributions

Comparisons of Original (MechMet), 28 mode representation (fit), and 28 mode reconstruction.

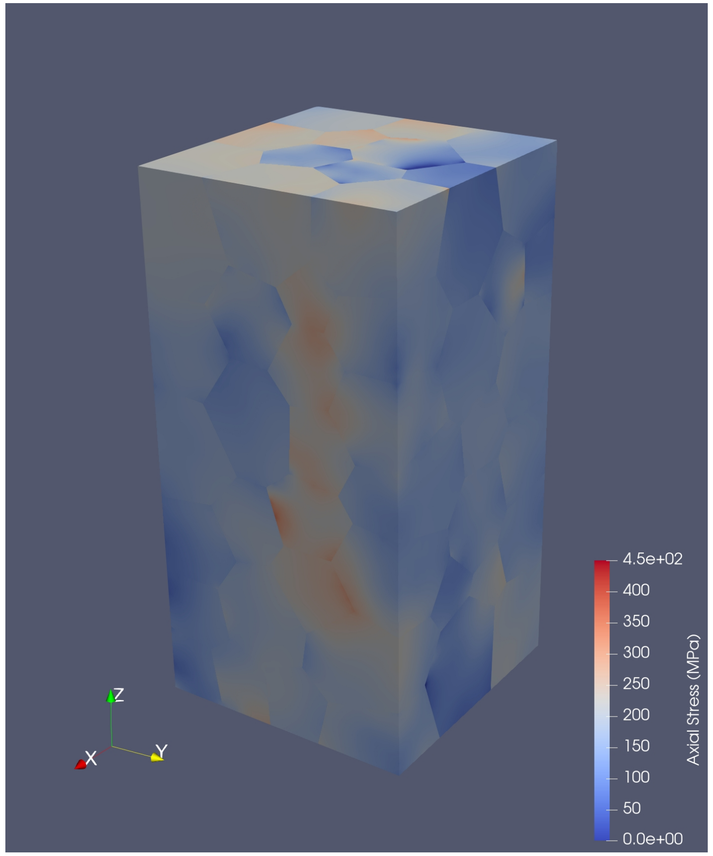
28Mode Representation



MechMet Solution



28Mode Reconstruction



$F_{28}/F_1 = 0.19$